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CYBERSECURITY

UNIVERSITÀ DEGLI STUDI DI TORINO



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Insegnamento: DeFi

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MODULE

Decentralized Finance — DeFi

Finance rebuilt as open, composable smart contracts — DEXs & AMMs, lending markets, oracles, stablecoins, liquid staking, RWAs, bridges & omnichain: features, mathematics, examples and risks

Blockchain and Distributed Ledger Technologies

Structured on Ch. 13 of “Mastering Ethereum” (2nd ed.), Antonopoulos, Wood et al. — CC BY-NC-ND

Learning objectives

By the end of this module you should be able to:

- 1 Explain how DeFi differs from traditional finance, and articulate its benefits and its honest limitations.
- 2 Describe the automated market maker model and DERIVE why the constant-product curve protects a liquidity pool.
- 3 Quantify impermanent loss and slippage on a concrete numeric example.
- 4 Explain overcollateralized lending, liquidation, keeper incentives, and how primitives COMPOSE (e.g. shorting).
- 5 Classify stablecoins, oracles, liquid staking, RWAs, bridges and omnichain protocols — with their specific risks.
- 6 Critically assess how decentralized DeFi really is, and name the main risk and regulatory challenges.

Outline

I	Foundations	What DeFi is, benefits, DeFi vs TradFi, composability
II	DEXs & AMMs	Order book → AMM; $x \cdot y = k$; the curve; slippage; impermanent loss
III	Lending markets	Collateral, liquidation, incentives, shorting, flash loans
IV	More primitives	Oracles, stablecoins, liquid staking, RWAs, bridges, omnichain
V	Critical view	(De)centralization, a risk matrix, regulation
VI	Examples & references	Significative protocols, takeaways, sources

PART I

Foundations

What DeFi is — and how it reimagines the financial system

What is Decentralized Finance?

DeFi is a complete financial ecosystem running entirely on-chain. Lending, borrowing, trading and investing are performed by **smart contracts** — with no banks, brokers or central authority. The protocol IS the institution, and its rules are public, immutable code.

Transparent & immutable

Every transaction is publicly recorded and verifiable; anyone can audit the state. This curbs fraud and removes the need to trust an operator.

Financially inclusive

A smartphone + internet grants access to credit, savings and investment — potentially reaching billions excluded from traditional banking.

Programmable & composable

Contracts combine like “money legos”, enabling instruments impossible in TradFi (e.g. collateral-free flash loans).

Honest caveat

Today's users are mostly speculators in wealthy nations, not the unbanked. Inclusion remains an aspiration, not yet the reality.

DeFi vs TradFi: getting a loan

TradFi — many gates

ID, provable address, national-insurance number, a bank account, signed papers, a good credit score — and it may still be DENIED, slowly. High barriers exclude billions.

DeFi — three things

A phone, an internet connection, and enough assets to OVERCOLLATERALIZE the loan. Once in place, the loan is instant, permissionless and decentralized.

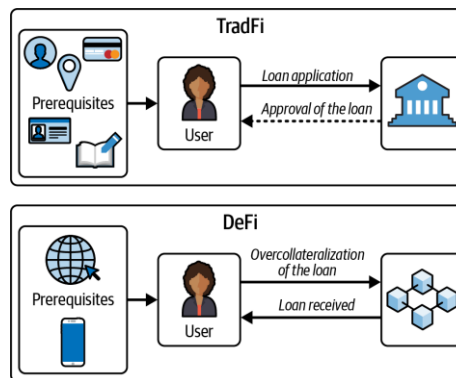


Figure 13-1 — traditional finance vs DeFi loan requirements. Source: Mastering Ethereum 2nd ed. (CC BY-NC-ND).

The openness trade-off & token acceptability

Permissionless openness is DeFi's superpower and its hardest problem: with no identity gate, TradFi mechanisms for credit-scoring and fraud resolution simply do not apply — which is why loans are overcollateralized rather than credit-assessed.

Each token is its own contract — “acceptability” matters

On the Arbitrum rollup, **USDC** (issued natively) and **USDC.e** (a bridged version) both target \$1 — yet they are DIFFERENT contracts.

- **Risk:** USDC.e inherits USDC's risk PLUS bridge risk (extra smart-contract exposure).
- **Acceptability:** some protocols accept only one, some both. “Same name, same price” does NOT mean “same money”.
- **Lesson:** in DeFi you must always check WHICH contract a token is before you trust or trade it.

PART II

Decentralized exchanges & AMMs

Trading with no intermediary — and the mathematics that makes it safe

From order books to automated market makers

A **decentralized exchange (DEX)** lets users trade peer-to-contract while keeping custody of their funds. How prices get set evolved through three stages:

Order book (on-chain)

Match buy & sell orders like a centralized exchange. EtherDelta (2017-18) tried it — but chains are slow and every order costs gas. Terrible UX; it never caught on.

AMM — Bancor

Pioneered the automated market maker: no order matching. Users deposit LIQUIDITY POOLS and a formula prices trades from the token ratio → continuous liquidity.

AMM — Uniswap

Popularized the AMM with the elegant constant-product formula $x \cdot y = k$. Anyone can become a market maker by supplying liquidity and earning fees.

The constant-product AMM: $x \cdot y = k$

A pool holds two tokens in quantities x and y . Every trade keeps the product $k = x \cdot y$ **constant**; the price of a token is simply the ratio of the reserves.

```
# Pool: 100 USDC and 10 ETH → price(ETH) = 100 / 10 = 10 USDC
# A trade shifts reserves to 200 USDC and 10 ETH:
price(ETH) = 200 / 10 = 20 USDC ← the ratio (and price) moved

swap output:  $\Delta y = (y \cdot \Delta x) / (x + \Delta x)$ 
```

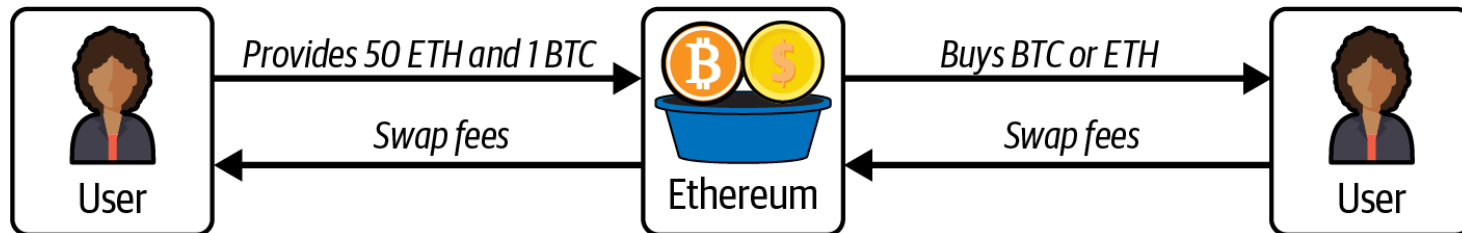


Figure 13-2 — a DEX liquidity pool. (“BTC” on a DEX is usually WBTC, a derivative, not native BTC.) Source: Mastering Ethereum 2nd ed. (CC BY-NC-ND).

LAB — DO IT NOW

Build a constant-product liquidity pool

Open the folder `lab-defi/` in Remix (VM). Complete `getAmountOut` and swap in `01-ConstantProductAMM_start.sol`, then reproduce the 1.98 and 66.67 outputs. The next slides explain the curve, slippage and impermanent loss you will observe.

0x625C19B1679fBA8b9693E97FcF8Ff06d1816372

MinimalERC20 – Token0 ETK

0x7784a30dDA6b6358414F93341C44f2076df73E67

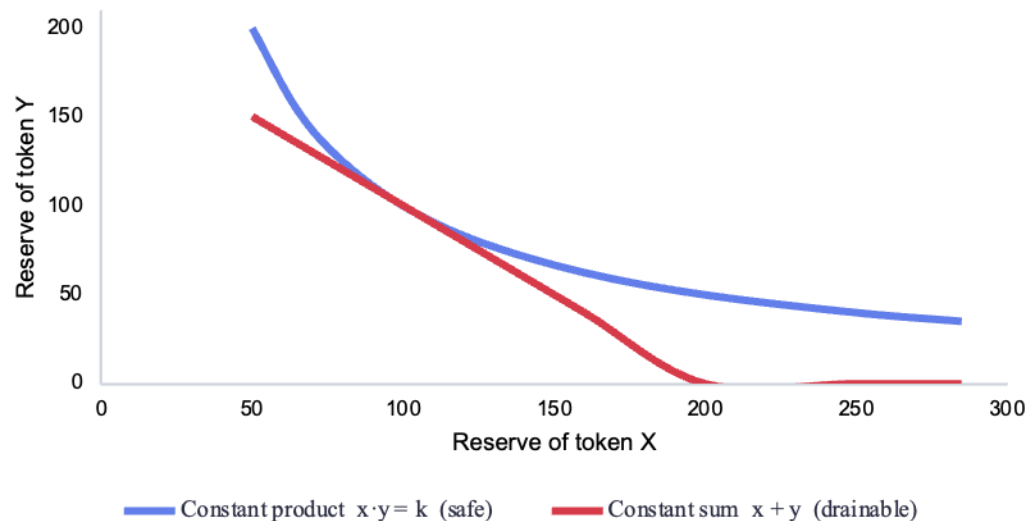
MinimalERC20 – Token1 USDT

0x0790d2041f72b4DD75Ac983C8f5358eCA73e92bC

ConstantProductAMM

Why a hyperbola? The curve that cannot be drained

Compare two pricing curves on ONE axis. The **constant-sum** line ($x + y = \text{const}$) touches both axes $\rightarrow$ a reserve can be traded to ZERO (drained). The **constant-product** hyperbola never touches an axis: as a token grows scarce its price rises toward infinity, so the pool is self-protecting.



Reads at a glance

Both curves meet at (100,100) — the same “spot” price. Away from it they behave oppositely.

Red line hits zero

At $x = 200$ the linear curve reaches $y = 0$: the pool is emptied. Fixed-price AMMs are unusable.

Blue curve stays up

The hyperbola's tail never reaches the axis — an economic barrier against complete drainage.

Slippage: big trades get a worse price

Because the curve steepens, the larger the trade relative to the pool, the worse the exchange rate. This gap between expected and executed price is SLIPPAGE — and it naturally discourages pool-draining trades.

Worked example — pool of 100 ETH and 200 USDC ($k = 20,000$):

```
swap 1 ETH → ≈ 1.98 USDC (close to the 1:2 spot ratio)
swap 50 ETH → 66.67 USDC (NOT 100 — a linear model would over-promise)
get all 200 USDC → requires infinite ETH ( $\Delta y \rightarrow y$  but never reaches it)
```

Small trade

Near-spot rate; slippage negligible. Deep pools = low slippage.

Large trade

Rate degrades sharply; the trader pays a premium set by the curve, not by any operator.

Impermanent loss — a numeric walk-through

By providing liquidity you are forced to sell the rising token and buy the falling one. If prices move a lot, your pool value can be LESS than simply holding — even after fees. It is “impermanent” only because it disappears if prices return.

```
# Deposit into an ETH/USDC pool.  ETH = $100.  
start:  10 ETH + 1000 USDC  →  k = 10,000,   deposit value = $2,000  
  
# ETH price DOUBLES to $200. Arbitrage rebalances the pool to keep k:  
new reserves ≈ 7.07 ETH + 1414 USDC  (√(k/p) and √(k·p))  
pool value  = 7.07·200 + 1414 = $2,828  
if you had just HELD = 10·200 + 1000 = $3,000  
impermanent loss = 3000 - 2828 = $172  (≈ 5.7%)
```

Rule of thumb: a 2× price move ⇒ ~5.7% IL; a 4× move ⇒ ~20%. Fees must exceed IL for LPing to pay.

Liquidity fragmentation & who provides liquidity

Anyone can be a market maker

Deposit a token pair and earn a share of trading fees proportional to your liquidity — no permission, no licence.

Fragmentation is the cost

Anyone can also launch a DEX, so liquidity is split across 100+ venues and many chains → thinner pools, worse slippage than one consolidated market.

Aggregators as a remedy

DEX aggregators (e.g. 1inch, CowSwap, Matcha) route a single swap across many pools to minimize slippage — a market response to fragmentation. Historical note: Uniswap was inspired by a 2016 Vitalik Buterin post and built in Vyper by Hayden Adams.

PART III

Lending markets

Algorithmic borrowing and lending without a bank

How a lending (money) market works

Lenders deposit assets into a pool and earn interest; borrowers draw from that pool against collateral. Smart contracts automate deposits, interest, collateral and repayment — no bank in the middle.

Lenders supply liquidity

Deposit crypto into the pool and earn interest. The deeper the pool, the more can be borrowed.

Algorithmic interest rates

Rates follow supply & demand: more borrowing demand → higher rates → attracts more lenders. Dynamic and market-driven.

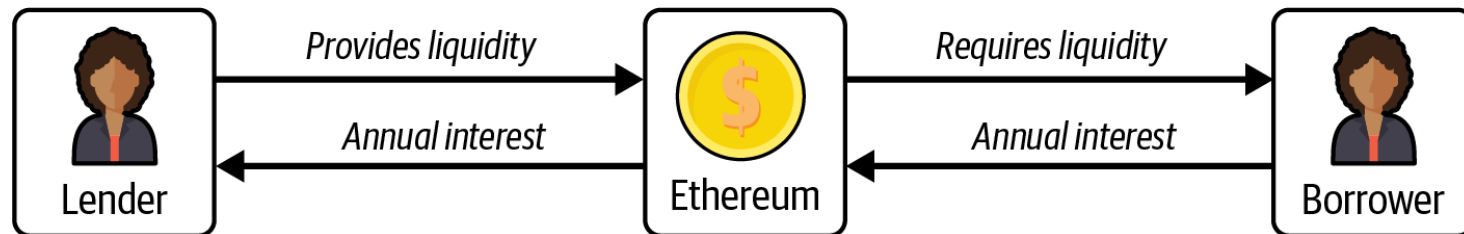


Figure 13-3 — basic lending-market flow: lender provides liquidity & earns interest; borrower withdraws it. Source: Mastering Ethereum 2nd ed. (CC BY-NC-ND).

Overcollateralization, liquidation & bad debt

Crypto is volatile, so loans are **overcollateralized**: at a 150% ratio, borrowing \$100 requires locking $\geq$ \$150 of collateral. The buffer absorbs price swings.

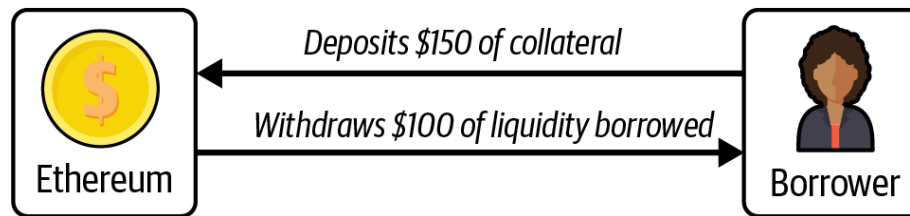


Figure 13-4 — a 150% collateralization ratio. Source: Mastering Ethereum 2nd ed. (CC BY-NC-ND).

Liquidation

If collateral value falls below a threshold, smart contracts sell it to repay the loan — protecting lenders and keeping the pool solvent.

Bad debt

If a sudden crash leaves collateral worth LESS than the debt, the shortfall is uncollectible — a loss to the protocol and its users.

Incentives & composability: building a short

Keepers & incentives: liquidations rely on anyone monitoring under-collateralized loans and triggering them — earning a slice of the liquidated collateral. Game-theoretic incentives keep the system solvent and self-sustaining.

Composability lets you re-create a **SHORT** (profit when a price falls) from a lending market + a DEX:

1. collateralize Asset A # deposit A in the lending market
2. borrow Asset B # the asset you expect to fall
3. sell B on a DEX # later: rebuy B cheaper, repay, keep the difference

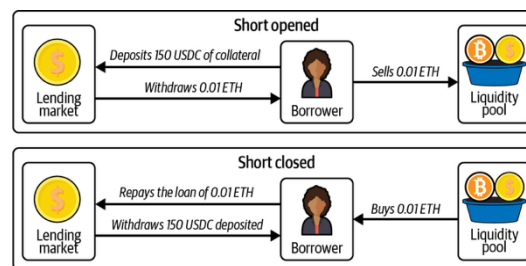


Figure 13-5 — a shorting strategy assembled from DeFi primitives. Source: Mastering Ethereum 2nd ed. (CC BY-NC-ND).

Flash loans — an instrument unique to DeFi

A **flash loan** lets you borrow ANY amount with NO collateral — on one condition: it must be borrowed AND repaid within the SAME transaction. If repayment fails, the entire transaction reverts as though it never happened.

Why it is possible

Atomicity: a transaction fully succeeds or fully reverts. The lender is risk-free because non-repayment unwinds everything.

Legitimate uses

Arbitrage across DEXs, collateral swaps, refinancing, self-liquidation — capital-efficient maneuvers impossible in TradFi.

Abuse vector

Amplifies price-oracle attacks: borrow huge, skew a pool's price, exploit a protocol that trusts that price, repay — all atomically.

The lesson

Never trust a manipulable spot price. Flash loans make any such weakness instantly and cheaply exploitable.

PART IV

More DeFi primitives

Oracles, stablecoins, liquid staking, RWAs, bridges & omnichain

Oracles in DeFi

Oracles bring off-chain data on-chain (covered fully in the Oracles module). In DeFi they are mission-critical — and a prime attack surface.

Price feeds

Lending markets need asset prices to value collateral and trigger liquidations.

Flash-loan attack risk

If a market reads price from a single DEX, a flash loan can skew it and force wrongful liquidations or under-priced borrows.

On-chain randomness

Deterministic Ethereum has no true randomness; oracles (e.g. VRF) supply it for lotteries, games, fair selection.

Decentralized by design

Good oracle networks are themselves decentralized — no single point of failure — so relying on them need not re-centralize DeFi.

Stablecoins — three designs, three risk profiles

A stablecoin targets a stable value (usually \$1), blending crypto's properties with fiat-like stability. The mechanism that holds the peg defines the risk:

Type	Backing	Example	Main risk
Fiat-collateralized	1:1 fiat reserves (custodian)	USDC, USDT	Trust in issuer & reserves
Crypto-collateralized	Overcollateralized crypto	DAI (Maker/Sky)	Collateral volatility
Algorithmic	None — supply algorithm	UST (failed)	Peg collapse / death spiral

A scaling limit: if a stablecoin's value on a chain exceeds the cost of attacking that chain, it creates an economic incentive to attack it — bounding how large a stablecoin can safely grow on a given network.

Real case — Terra / LUNA collapse (May 2022)

The definitive algorithmic-stablecoin failure: about \$45B erased in three days — a live demonstration of the “death spiral” from the previous slide.

The design

UST was an ALGORITHMIC stablecoin: 1 UST was always mint/redeemable for \$1 of its sister token LUNA. Demand was propped up by Anchor paying ~20% yield on UST.

The trigger (7-9 May)

Large UST withdrawals from Anchor and big UST sells on Curve broke the \$1 peg. Confidence cracked and a bank run began.

The death spiral

Arbitrageurs redeemed UST for newly-minted LUNA and sold it. LUNA supply exploded from ~1B to ~6.5T tokens; its price fell from ~\$80 to almost zero.

Lesson

An algorithmic peg with no real collateral is only as strong as confidence. When it breaks, minting to defend the peg hyperinflates the sister token — the exact risk on the stablecoin slide.

Liquid staking & real-world assets

Liquid staking

Stake ETH yet stay liquid: receive a derivative token (e.g. stETH from Lido) that keeps earning rewards but can be traded or reused in DeFi.

Extra risks

The derivative is a peg (like a stablecoin) PLUS slashing risk if validators misbehave. Restaking (EigenLayer) layers on further, debated, risk.

Systemic concentration

Lido has held ~29% of staked ETH; beyond 33% a single staker could threaten finalization. A 2022 self-cap vote did NOT pass.

Real-world assets (RWA)

Tokenized bonds, real estate, commodities, equities → fractional ownership & access to risk-free rates — but require a CUSTODIAN, in tension with trustlessness.

Bridges: moving assets between chains

Chains are isolated; bridges connect them so tokens and data can move (e.g. Ethereum → BSC). Most are CENTRALIZED — a party approves and unlocks liquidity — which is why bridges are among the most-hacked DeFi protocols.

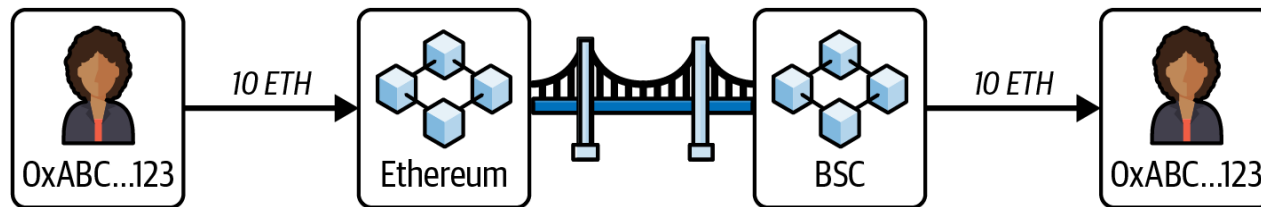


Figure 13-6 — a bridge connecting two blockchains. Source: Mastering Ethereum 2nd ed. (CC BY-NC-ND).

Lock & mint

Lock tokens on A, mint a wrapped receipt on B; burn on B to unlock on A.

Burn & mint

Project controls the token: burn on A, mint an equal amount on B — supply preserved.

Liquidity bridge

Pools on both chains; deposit on A, receive from B's pool. No new tokens minted (most common).

Omnichain protocols

Omnichain (cross-chain messaging) protocols go beyond moving tokens: they let smart contracts on different chains COMMUNICATE, so a single action can span ecosystems — e.g. start a swap on Ethereum and receive the funds on BSC.

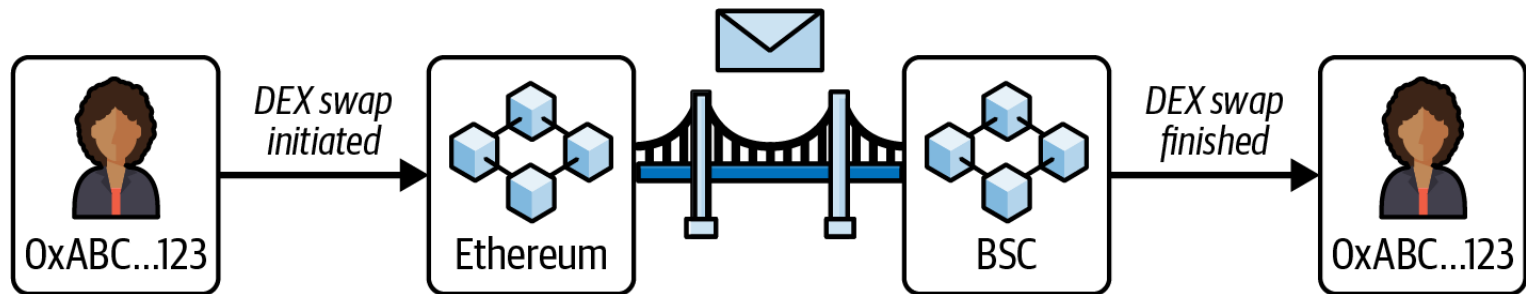


Figure 13-7 — an omnichain messaging protocol (swap initiated on one chain, settled on another). Source: Mastering Ethereum 2nd ed. (CC BY-NC-ND).

Examples (from the Oracles module): LayerZero, Wormhole, Chainlink CCIP, Circle CCTP. They aim for a unified interoperability layer across many chains.

PART V

A critical view

How decentralized is DeFi really — and what can go wrong

(De)centralized finance — the honest picture

“Decentralization” is often an aspiration more than a fact. Most DeFi protocols still rely on privileged admin addresses or a core team's decisions — a concentration of power inside a supposedly decentralized system.

Why some centralization helps early

A young ecosystem benefits from fast decisions and quick fixes; a core team provides the agility to adapt and grow.

The intended trajectory

“Progressive decentralization” — reduce privileged control over time as the protocol and its community mature.

Early adopters carry real responsibility: balancing today's need for efficiency against the founding vision of openness, transparency and inclusivity.

A risk matrix: every primitive has its hazard

Beyond generic smart-contract risk (bugs, malicious/careless custodians), each primitive carries a SPECIFIC economic risk:

Primitive	Characteristic economic risk
DEX / liquidity provision	Impermanent loss; high slippage in thin pools
Lending market	(Wrongful) liquidation; bad debt on volatile collateral
Stablecoin	Loss of the peg (esp. algorithmic designs)
Liquid staking	Peg loss + slashing + systemic concentration
Bridge	Custodial compromise — the most-hacked category
Oracle-dependent protocol	Price manipulation, amplified by flash loans

Users must understand every risk of a protocol BEFORE using it — even experts are routinely caught out.

Regulation & sustainability challenges

Regulation targets people, not code

Regulators cannot stop an immutable contract, so they pursue DEVELOPERS and USERS. Tornado Cash's developer was jailed; Samourai Wallet's team faced action. CEXs flag “tainted” addresses.

Liquidity fragility

Many protocols work only while usage is INCENTIVIZED; liquidity evaporates in downturns. Real product-market fit is still limited.

Copy-paste “vampire” forks

Many protocols merely fork incumbents to lure liquidity without real innovation — diluting the ecosystem.

An uncertain but promising future

DeFi shows genuine product-market fit in places; it may become global infrastructure — or stay a niche. Early adopters shape which.

PART VI

Examples & references

Significative protocols, takeaways and sources

Significative DeFi protocols by primitive

A non-exhaustive, illustrative map (the ecosystem moves fast):

DEX / AMM

Uniswap, Curve (stable-swaps), Balancer, Bancor (pioneer)

Stablecoins

USDC, USDT (fiat) · DAI / USDS (crypto) · UST (algo, failed)

Oracles

Chainlink, Pyth, RedStone — price feeds & VRF

Real-world assets

Tokenized treasuries / bonds, real-estate fractions

Lending markets

Aave, Compound, MakerDAO / Sky (CDP)

Liquid staking

Lido (stETH), Rocket Pool · EigenLayer (restaking)

Bridges / omnichain

LayerZero, Wormhole, Chainlink CCIP, Circle CCTP

Aggregation & privacy

1inch, CowSwap (DEX aggregators) · Tornado Cash (mixer)

“DeFi legos”: these primitives compose — lending + DEX = shorting; stablecoins deepen DEX liquidity; oracles secure lending.

Key takeaways

- **DeFi = finance as composable smart contracts** — Permissionless, transparent, no intermediary — the protocol is the institution.
- **AMMs replaced order books** — $x \cdot y = k$ makes a pool self-protecting; LPs face impermanent loss, traders face slippage.
- **Lending runs on overcollateralization** — No credit scores; liquidation + keeper incentives keep pools solvent; flash loans need none.
- **Primitives compose into “legos”** — Lending + DEX = shorting/longing; oracles, stablecoins & bridges glue it together.
- **Trust still leaks in** — Oracles, RWA custodians, admin keys, centralized bridges — “decentralized” is a spectrum and a goal.
- **Know the risks** — Smart-contract bugs, per-primitive economic risk, liquidity fragility, and a hostile regulatory frontier.

References

- [1] Antonopoulos A., Wood G. et al. — Mastering Ethereum, 2nd ed., Ch. 13 “Decentralized Finance” (CC BY-NC-ND). PRIMARY SOURCE. masteringethereum.xyz · github.com/ethereumbook/ethereumbook
- [2] Adams H. et al. — Uniswap V2 / V3 whitepapers & docs (constant-product AMM). docs.uniswap.org
- [3] Aave & Compound protocol documentation (lending markets, liquidation). docs.aave.com · docs.compound.finance
- [4] MakerDAO / Sky — DAI / USDS crypto-collateralized stablecoin. docs.sky.money
- [5] Circle (USDC) & Tether (USDT) — fiat-collateralized stablecoins. circle.com · tether.to
- [6] Lido (stETH) & EigenLayer — liquid staking & restaking. docs.lido.fi · docs.eigenlayer.xyz
- [7] Chainlink — decentralized oracles, price feeds & VRF. docs.chain.link
- [8] LayerZero, Wormhole, Chainlink CCIP, Circle CCTP — bridges & omnichain (project docs).
- [9] On flash loans & price-oracle manipulation — DeFi security literature (see the Smart-Contract Security module).
- [10] TerraUSD (UST) collapse analyses; bridge-hack post-mortems (Ronin, Wormhole) — public incident reports.

Figure credits & licence

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All explanatory text, the worked examples, and the constant-product chart are original course material produced for Blockchain & DLT.

Questions?

DeFi rebuilds finance as open, composable code — a genuine innovation that still carries new, poorly-understood risks. Use it with eyes open.

Grazie per l'attenzione



L'Europa investe sul Piemonte, il Piemonte investe su di te